(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

THIRD SEMESTER [B.TECH] JANUARY 2024

Paper Code: CIC209 ***Subject: Data Structures***

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt five questions including Q.No.1 which is compulsory. Select One question from each unit. Assume missing data, if any.

Q1 Attempt the following: **(5x3=15)**

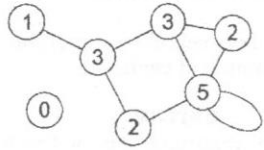
a) What do you understand by underflow and overflow in stack and queue data structures?

b) Why is it required to analyze the time complexity and space complexity of an algorithm?

c) Write short note on Bucket Hashing.

d) Write an algorithm to count the number of nodes of a binary tree.

e) Represent the following graph using Adjacency Matrix and Adjacency List representations:



UNIT-I

Q2 a) Convert the following infix expression into postfix expression using Stack: **(5)**

(A + B) / C * (D – E)^F – G

b) Write an algorithm to insert an element in the sorted doubly linked list. **(5)**

c) Write short note on double ended queue (deque). Can we implement stack and queue data structures using deque? Comment. **(5)**

Q3 a) What are the different classifications of data structures? Explain briefly all types of data structures. **(5)**

b) Consider a two-dimensional array or integers A [-11..10, 9..20]. Each element requires 4 bytes of memory storage. If the base address of the Array A is 100. You are required to derive the formula to locate the address of each element then determine the location (i.e. memory address) of A[5][15]. **(5)**

c) Explain similarities and dissimilarities between stack and queue data structures. Also list two applications of each. **(5)**

UNIT-II

Q4 a) Step by step construct the binary tree based on the below given traversals of the tree: **(5)**

Preorder: A B D C E G F H I

Inorder: B D A G E C H F I

b) What is AVL Tree? Explain different types of rotations used in AVL Tree. **(5)**

P.T.O.

CIC-209

P-1/3

(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

Third Semester, November 2022

Paper Code: CIC-209 **Subject: Data Structures**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1	**Attempt the following:**		
(a)	Briefly explain Abstract Data Type model with the help of a diagram.	2	CO1
(b)	Why are data structures required? List various types of data structures.	2	CO1
(c)	What is sparse matrix? Explain briefly the 3-tuple (or triplet) representation of sparse matrix with an example.	2	CO2
(d)	Differentiate between static implementation (using arrays) and dynamic implementation (using pointers) of a list data structure.	2	CO2
(e)	Define the terms depth, height, internal node, and leaf node with respect to binary trees with the help of an example.	2	CO2
2 (a)	Write algorithms for: (i) insertion of an element in a sorted singly linked list, and (ii) deletion of an element from a sorted singly linked list.	5	CO2
(b)	Consider the following Preorder and Inorder Traversals of a binary tree: Preorder : A B D E C F Inorder : D B E A F C You are required to construct the binary tree based on above given traversals. Show all the steps to construct the tree.	5	CO2
3 (a)	What is binary search tree? Write the algorithms for Preorder, Inorder and Postorder Traversals of the tree.	5	CO2
(b)	Why circular queue preferred over simple queue? Demonstrate and explain with the help of a suitable example.	5	CO2
4 (a)	Convert the following infix expression into postfix expression using stack: A + B * C – D / (E * F) ^ G	5	CO2
(b)	Construct the binary search tree using the following data elements: 35, 70, 50, 10, 48, 80, 55, 11, 85 Show binary search tree after insertion of each data element.	5	CO2

UNIT-III

Q6. a) What information is contained by the Symbol Table? Explain the contents and its capabilities of symbol table. **(7)**

b) What is error? Explain sources of errors and explain various types of errors at each phase of compiler. **(5.5)**

Q7. a) Describe various data structures of symbol tables in detail. **(5.5)**

b) What are the different storage allocation strategies in the runtime environment of the compiler? Explain. **(7)**

UNIT-IV

Q8. a) What is code generation? Explain various issues of Code Generation. **(6.5)**

b) Briefly explaina) code generation from DAGs b) Value number and algebraic laws **(6)**

Q9. a) How Peep Hole Optimization is useful in Code Optimization& Code Generation phase. Explain briefly. **(5.5)**

b) What are basic blocks and Flow Graphs? Explain with the help of examples. **(7)**

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